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Bond Market Resiliency: The Role of Insurers

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1 Big picture

- **Question.** Who provides stabilizing liquidity in corporate bond markets when dealers and open-end funds are under stress, and what role do insurance companies play in market resiliency?
- **Setting.** The paper studies the U.S. corporate bond market around the COVID-19 liquidity crisis, especially the two-week crisis window from March 6 to March 19, 2020, just before the Federal Reserve announced major corporate bond market facilities.
- **Core idea.** Dealers are usually buyers of “first resort”: they smooth short-term order-flow imbalances, but they may not have enough balance sheet capacity or incentives to absorb a market-wide sell-off. A resilient market therefore needs long-horizon investors who can buy when temporary price dislocations appear.
- **Candidate stabilizers.** Insurance companies are natural candidates because they are large corporate bond holders, have long-term liabilities, and may have stable funding from insurance underwriting. At the end of 2019, U.S. insurers held roughly 30% of outstanding corporate bonds.
- **Main empirical result.** During the March 6–19 crisis window, dealers’ cumulative net purchases fell by about \$5 billion, while insurers were net buyers of about \$2.5 billion in bonds.
- **Funding-stability result.** Insurers with lower variation in insurance funding were more likely to be net buyers during the crisis. A one-standard-deviation decrease in funding variation raises the probability of being a crisis-period net buyer by about 3.1 percentage points, roughly 9% of the unconditional 35% probability.
- **Relationship-trading result.** Insurers bought more from, and sold less to, dealers with whom they had stronger prior trading relationships. Dealers later rewarded supporting insurers with better execution quality.
- **Market-liquidity result.** Dealers with relationships to more stable-funded insurers charged lower transaction costs, especially when bonds were exposed to mutual-fund fire-sale pressure.
- **Aggregate-price result.** For the median investment-grade bond, yields would have been about 25 basis points higher during the crisis absent ownership by stable-funded insurers.
- **Economic takeaway.** Insurers can be stabilizers rather than amplifiers when they are not directly hit by the shock. Their liability-side funding stability can support asset-side liquidity provision and improve bond market resiliency.

2 Data and measurement (key objects)

2.1 Main transaction data

- **NAIC insurer transaction data.** The primary data source is insurers’ corporate bond transactions reported quarterly to the National Association of Insurance Commissioners (NAIC). The sample covers life and property-and-casualty (P&C) insurers from 2017 to 2020.
- The NAIC Schedule D filings contain bond acquisitions, dispositions, and positions acquired and disposed of within the same year.

- The data identify the bond, issuer, execution date, dollar amount, par amount, buy/sell direction, insurer identity, and dealer counterparty.
- This two-sided counterparty information is crucial: it lets the authors connect insurer balance-sheet characteristics, insurer-dealer relationships, and dealers' liquidity provision.
- **Final NAIC trade sample.** After excluding primary-market trades and cleaning incomplete or nonapplicable observations, the sample contains 768,444 secondary-market bond trades, 15,763 bonds, and 2,879 corporate issuers.

2.2 Bond, insurer, dealer, and mutual fund data

- **Bond characteristics.** Mergent FISD provides issue date, maturity date, par amount outstanding, credit ratings, coupon rates, and lead underwriters.
- **Insurer financial data.** S&P Global Market Intelligence provides quarterly statutory cash-flow and balance-sheet data. The authors use these data to measure insurers' funding stability, assets, surplus, liabilities, cash holdings, risk-based capital ratios, and bond-portfolio ratings.
- **Final insurer sample.** Insurers must have five years of pre-2020 balance-sheet data, trade at least once per year, have an RBC ratio above 200%, and satisfy size/solvency filters. The final sample contains 1,647 insurers: 465 life insurers and 1,182 P&C insurers.
- **TRACE dealer data.** The regulatory version of TRACE provides secondary-market corporate bond transactions and dealer identities. These data are used to measure all customer-dealer trades and transaction costs.
- **Mutual fund exposure.** Refinitiv eMaxx gives institutional bond holdings, including mutual fund holdings. CRSP mutual fund holdings plus Morningstar daily fund flows are used to compute imputed mutual fund outflow pressure at the bond-day level.
- **Dealer characteristics.** SEC FOCUS reports provide dealer-level balance sheet and income statement variables such as dealer size, leverage, and ROA.

2.3 Key periods

- **Precrisis period.** February 21 to March 5, 2020.
- **Crisis period.** March 6 to March 19, 2020. This is the window in which corporate bond markets were under severe stress before the Fed's major corporate bond market facilities.
- **Postcrisis execution-quality period.** April to December 2020, used to study whether dealers rewarded insurers that supported them during the crisis.
- **Lookback period for relationships.** 2017–2019, used to define prior insurer-dealer trading relationships.

2.4 Funding stability

- The key insurer characteristic is **variation in insurance funding** (VIF). Lower VIF means more stable funding.

- Insurance funding is based on cash flows from underwriting plus net financing flows, scaled by lagged liabilities.
- The main VIF measure is the five-year historical standard deviation of this insurance funding ratio through 2019:Q4.
- The paper also uses alternative measures: underwriting-profitability variation, high-VIF indicators, exponentially weighted VIF, first-differenced VIF, semideviation-based VIF, and semideviation of first-differenced funding.
- Life insurers are generally larger and more leveraged than P&C insurers, while P&C insurers have a more pronounced increase in net-buyer rates from the precrisis to the crisis window.

2.5 Dealer relationships and transaction costs

- An insurer-dealer relationship is measured by the share of an insurer’s past trading volume conducted with a given dealer over 2017–2019.
- A dealer’s connected-insurer VIF is measured as the volume-weighted average VIF of the insurers with which that dealer traded in the lookback period.
- Transaction costs are measured using customer-dealer trade prices in TRACE relative to the most recent interdealer benchmark price.
- Trade-size categories are micro (\$1–\$100,000), odd-lot (\$100,000–\$1,000,000), round-lot (\$1,000,000–\$5,000,000), and block (above \$5,000,000).

2.6 Mutual fund fire-sale exposure

- The paper uses two main proxies for mutual-fund selling pressure:
 - **Mutual fund holdings:** the log of one plus the par amount of a bond held by mutual funds as of 2019:Q4.
 - **High imputed mutual fund outflows:** an indicator equal to one when a bond’s imputed mutual fund flow is negative and falls in the top quintile of outflows.
- These variables ask whether insurers bought the bonds most likely to be dumped by open-end funds during the crisis.

3 Models and empirical specifications (all equations + interpretation)

3.1 Underwriting profitability and insurance funding

The baseline underwriting-profitability ratio is

$$UP_{i,t} = \frac{Premiums_{i,t} - Losses_{i,t} - Expenses_{i,t}}{Liabilities_{i,t-1}}.$$

The paper’s broader insurance-funding ratio is

$$IF_{i,t} = \frac{NetCashFromUnderwriting_{i,t} + NetCashFromFinancing_{i,t}}{Liabilities_{i,t-1}}.$$

The main funding-instability measure is

$$VIF_i = SD_{2015Q1:2019Q4}(IF_{i,t}).$$

Interpretation.

- Low VIF_i means the insurer's noninvestment funding is stable.
- Stable funding should allow the insurer to buy illiquid bonds during temporary price dislocations.
- The paper treats lower VIF_i as the main source of insurers' ability to act as liquidity suppliers during the crisis.

3.2 Insurer net-buyer model

The first main empirical model is a linear probability model:

$$NetBuyer_{i,t} = \alpha_1 VIF_{i,2019Q4} + \alpha_2 Crisis_t \times VIF_{i,2019Q4} + X'_{i,2019Q4} \gamma + Crisis_t \times X'_{i,2019Q4} \delta + \lambda_{type(i),t} + \varepsilon_{i,t}.$$

Here:

- $NetBuyer_{i,t} = 1$ if insurer i 's aggregate net purchases are positive in period t ;
- t indexes the precrisis or crisis period;
- $Crisis_t = 1$ during March 6–19, 2020;
- X_i includes total assets, asset growth, RBC ratio, leverage, cash-to-assets, and bond-portfolio rating;
- $\lambda_{type(i),t}$ are insurer-type-period fixed effects.

Interpretation.

- The coefficient of interest is α_2 .
- If stable funding matters more during stress, then $\alpha_2 < 0$: high VIF insurers are less likely to buy during the crisis.
- The paper finds exactly this. A one-standard-deviation decrease in funding variation increases the probability of being a crisis-period net buyer by about 3.1 percentage points.

3.3 Relationship trading with dealers

To test whether insurers supported relationship dealers, the paper estimates

$$ShareBuyFromDealer_{i,j,t} = \beta_1 Crisis_t + \beta_2 SharePastTrading_{i,j} + \beta_3 Crisis_t \times SharePastTrading_{i,j} + X'_{i,2019Q4} \Gamma +$$

A parallel regression replaces the dependent variable with $ShareSellToDealer_{i,j,t}$.

Interpretation.

- $SharePastTrading_{i,j}$ is the average monthly share of insurer i 's trading volume with dealer j over 2017–2019.
- The relationship-support hypothesis predicts $\beta_3 > 0$ for buys and $\beta_3 < 0$ for sells.

- This is what the paper finds: during the crisis, insurers increased purchases from relationship dealers and reduced sales to them.
- The asymmetry matters: insurers were not just using familiar dealers for all trades; they tilted trading in a way that helped those dealers absorb inventory imbalances.

3.4 Transaction-cost measure

For a customer-dealer trade k , transaction cost is measured as

$$Cost_k = \ln \left(\frac{TradePrice_k}{BenchmarkPrice_l} \right) \times TradeDirection_k,$$

where $BenchmarkPrice_l$ is the price of the most recent interdealer trade and $TradeDirection_k = 1$ for a customer buy and -1 for a customer sell. The value is multiplied by 10,000 to express costs in basis points.

Interpretation.

- The formula signs customer buys and sells so that higher $Cost_k$ means a worse transaction cost for the customer.
- Comparing customer-dealer prices with interdealer benchmark prices gives a transaction-level measure of dealer markups/markdowns.

3.5 Dealer liquidity provision and connected-insurer funding

The paper estimates

$$Cost_k = \theta_1 Crisis_t + \theta_2 VIF_j + \theta_3 Crisis_t \times VIF_j + Z'_{b,t} \Pi + \phi_r + \phi_d + \phi_s + \varepsilon_k,$$

where:

- VIF_j is the volume-weighted average funding variation of dealer j 's relationship insurers;
- $Z_{b,t}$ includes bond age, time to maturity, and outstanding amount;
- ϕ_r , ϕ_d , and ϕ_s are rating, trade-direction, and trade-size fixed effects.

The more stringent versions use bond-day-trade-size-trade-direction fixed effects and dealer controls.

Interpretation.

- If relationships with stable-funded insurers improve dealer liquidity provision, then $\theta_3 > 0$: dealers connected to less stable-funded insurers charge higher costs during the crisis.
- Because lower VIF means more stable funding, a positive coefficient on VIF_j means worse liquidity when the dealer's insurance-client network is less stable.
- The paper finds that a one-standard-deviation reduction in dealer connected-insurer VIF lowers transaction costs by about 13 basis points during the crisis, or about 9 basis points with strong fixed effects.

3.6 Stable-funded insurer holdings and counterfactual yields

To study aggregate price support, the paper first estimates insurer purchases of familiar issuers:

$$NetBuys_{i,b,f,t} = \eta_1 IssuerHoldings_{i,f,2019Q4} + \eta_2 Crisis_t \times IssuerHoldings_{i,f,2019Q4} + Controls'_{i,b,t} \eta + Crisis_t \times Controls_{i,b,t} \eta$$

Then it estimates time-varying yield loadings on stable-funded insurer holdings:

$$Yield_{b,f,t} = \alpha_t + \beta_t StableFundedInsurerHoldings_{f,2019Q4} + \varepsilon_{b,f,t}.$$

Interpretation.

- Insurers tend to buy bonds from issuers they already hold, likely because familiar issuers require less information acquisition during a fast-moving crisis.
- The counterfactual sets stable-funded insurer holdings to zero and asks how high yields would have been without that ownership support.
- The paper finds a striking effect for investment-grade bonds during the crisis: the median investment-grade yield would have been about 25 basis points higher absent stable-funded insurer ownership.

3.7 Mutual fund fire-sale exposure

The bond-day model is

$$InsurerBuys_{b,t} = \rho_1 MFHoldings_{b,2019Q4} + \rho_2 Crisis_t \times MFHoldings_{b,2019Q4} + Controls'_{b,t} \rho + \psi_r + \psi_t + \varepsilon_{b,t}.$$

The same model is also estimated using insurer sales and using high imputed mutual fund outflows:

$$HighMFOutflows_{b,t} = 1 \quad \text{if bond } b \text{ is in the top quintile of negative imputed mutual fund flows on day } t.$$

Finally, the paper links mutual-fund exposure to transaction costs with a triple-interaction model:

$$Cost_k = \gamma_1 Crisis_t + \gamma_2 VIF_j + \gamma_3 MFShare_{b,2019Q4} + \gamma_4 Crisis_t \times VIF_j + \gamma_5 VIF_j \times MFShare_{b,2019Q4} + \gamma_6 Crisis_t \times VIF_j \times MFShare_{b,2019Q4} + Controls + FE + \varepsilon_k.$$

Interpretation.

- The buy-side result asks whether insurers bought bonds exposed to mutual-fund fire sales.
- The cost-side result asks whether dealers connected to stable-funded insurers provided especially cheap liquidity in bonds exposed to mutual-fund fire sales.
- The paper finds yes to both: insurers bought more exposed bonds, and stable-funded insurer relationships mattered most where mutual-fund fragility was likely largest.

4 Identification and empirical credibility (what makes the estimates credible)

4.1 What is hard to identify

- The key empirical challenge is that insurers may buy during the crisis for many reasons: stable funding, better capitalization, larger size, existing positions, or dealer relationships.
- Dealers with stable insurer clients may also differ from other dealers in size, market share, balance-sheet capacity, underwriting roles, or client composition.
- Bonds exposed to mutual funds may differ from other bonds in liquidity, rating, maturity, issue size, and issuer quality.
- The paper cannot randomly assign insurers to funding structures or dealers to insurance-client networks, so credibility comes from high-dimensional controls, fixed effects, and careful cross-sectional comparisons.

4.2 What identifies insurer stabilizing behavior

- Aggregate evidence compares dealers' and insurers' net purchases during the same crisis window.
- Cross-sectional evidence compares insurers with different funding stability, controlling for insurer type and balance-sheet characteristics.
- The key identifying variation is whether insurers with more stable historical insurance funding bought more precisely during the crisis, not merely whether they usually buy more.
- Robustness checks use alternative precrisis windows, maturing-bond adjustments, insurer-group clustering, size-weighted regressions, and alternative funding-stability measures.

4.3 What identifies relationship-based liquidity support

- The paper uses prior insurer-dealer trading over 2017–2019 to measure relationships before the crisis.
- By comparing the same insurer's buying and selling shares across dealers during the same period, the paper tests whether crisis trading tilts toward relationship dealers.
- The inclusion of insurer-day and dealer-day fixed effects helps control for time-varying insurer and dealer shocks.
- The nonunderwriter-dealer tests reduce the concern that the result is driven by dealers' underwriting obligations rather than trading relationships.

4.4 What identifies effects on transaction costs

- TRACE provides all customer-dealer trades and dealer identities, allowing the paper to connect transaction costs to a dealer's relationship-insurer funding stability.
- Bond-day-trade-size-trade-direction fixed effects compare trades in the same bond on the same day and with similar size/direction.

- Dealer controls and dealer-day fixed effects mitigate concerns that large or healthier dealers both attract stable insurers and charge lower transaction costs.
- The strongest evidence comes from the interaction with the crisis window and mutual-fund selling pressure: connected stable insurers matter most exactly where liquidity pressure is largest.

4.5 Limitations

- The evidence is observational; the paper does not randomly assign insurers’ funding stability or trading relationships.
- The COVID-19 shock is special: insurers were relatively healthy and not directly hit by large liability-side stress. In other shocks, insurers could instead amplify selling pressure.
- The counterfactual yield exercise is informative but not a fully structural general-equilibrium counterfactual.
- The analysis focuses on corporate bonds and insurance companies, so other stable-funded investors may play similar roles in other fixed-income markets.

4.6 Relation to the literature

- **Bond market fragility.** The paper extends work on dealer balance-sheet constraints, mutual-fund runs, and COVID-19 bond-market stress by identifying a stabilizing investor group.
- **Insurance companies.** The paper adds to the insurance literature by showing how liability-side funding stability supports asset-side liquidity provision.
- **OTC relationship trading.** Existing work studies how investors benefit from dealer relationships; this paper studies how dealers benefit from relationships with stable investors.
- **Investor composition and liquidity.** The paper shows that market liquidity depends not just on dealers but also on the funding structures and trading incentives of final investors.

5 Tables and figures

5.1 Table 1 and Figure 1: aggregate market stress and insurer buying

Read:

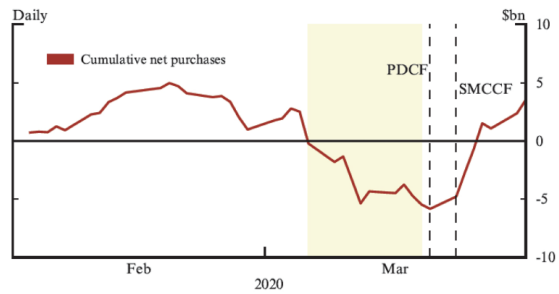
- Table 1 shows that insurers’ balance-sheet conditions changed little from 2019Q4 to 2020Q1. This matters because insurers were not themselves suffering a severe funding shock at the start of the crisis.
- The sample includes 1,647 insurers, split into 465 life insurers and 1,182 P&C insurers.
- Figure 1 is the basic market-resiliency fact. Dealers’ cumulative net purchases fell by about \$5 billion during March 6–19, while insurers bought about \$2.5 billion on net.
- The insurer net buying is economically meaningful: it represents roughly half of dealers’ net selling over the same crisis period.

Table 1
Insurers' financial conditions around the COVID-19 crisis

	All insurers		Life insurers		P&C insurers	
	2019Q4	2020Q1	2019Q4	2020Q1	2019Q4	2020Q1
Number of firms	1,647	1,647	465	465	1,182	1,182
Total assets	5,321	5,250	15,507	15,330	1,314	1,308
Surplus	620	612	965	991	485	464
Total liabilities	4,681	4,610	14,508	14,288	814	824
Leverage	59.5	60.3	78.1	78.6	52.2	53.1
Cash-to-assets	2.3	2.6	1.7	1.9	2.5	2.8
Bond portfolio rating	1.3	1.3	1.4	1.4	1.2	1.2

This table reports how financial conditions changed around the COVID-19 crisis for the insurance companies in our sample. We obtain quarterly balance sheet data from S&P Global Market Intelligence for 2019Q4 and 2020Q1 for the 1,647 insurers in our sample. The column *All insurers* reports summary statistics for the full sample, while the *Life* and *Property & Casualty* columns present type-specific statistics. The insurers are compared across the following metrics: *Total assets*, in millions of dollars. *Surplus*, in millions of dollars. *Total liabilities*, in millions of dollars. *Leverage*, defined as one minus the ratio of equity to total assets, reported in percentages. *Cash-to-assets*, the ratio of cash holdings to total assets, reported in percentages. *Bond portfolio rating*, the insurer's weighted average NAIC bond portfolio rating (1=highest through 6=lowest).

A Dealers' cumulative net purchases



B Insurers' cumulative purchases

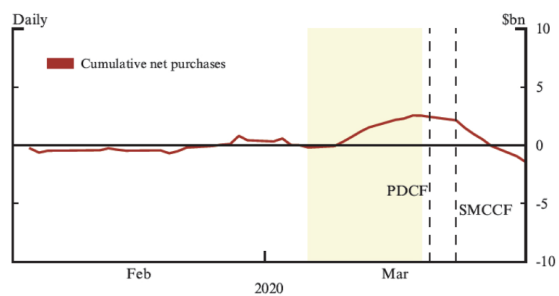


Figure 1
Dealers' and insurers' cumulative net purchases in the secondary corporate bond markets during the COVID-19 crisis

Panel A shows changes in dealers' cumulative net purchases from 2/1/2020 to 3/31/2020. Dealers' cumulative net purchases are defined as the difference between cumulative dealer buys and sells in the secondary corporate bond market, calculated using TRACE data. Panel B covers the same period and presents insurers' cumulative net purchases of bonds issued before 2/1/2020 and traded in the secondary corporate bond market. Insurer cumulative net purchases are defined as the difference between cumulative insurer buys and sells, calculated using NAIC data. The shaded areas mark the crisis period from 3/6/2020 to 3/19/2020. *Source: Authors' creation, using data from TRACE and NAIC.*

- The figure motivates the rest of the paper: if dealers were selling, then some nondealer investors had to absorb the selling pressure.

5.2 Table 2, Table 3, and Table 4: funding stability and insurer net buying

Table 2 Insurer characteristics before the COVID-19 crisis																				
	All insurers (n=1,647)						Life insurers (n=465)						P&C insurers (n=1,182)							
	Mean	Std. dev.	Q1	Median	Q3	Mean	Std. dev.	Q1	Median	Q3	Mean	Std. dev.	Q1	Median	Q3	Mean	Std. dev.	Q1	Median	Q3
log(total assets)	12.83	2.07	11.3	12.57	14.08	13.99	2.45	12.17	13.78	15.82	12.38	1.7	11.09	12.16	13.42					
Asset growth	0.04	0.11	0.0	0.04	0.08	0.04	0.11	0.0	0.03	0.07	0.04	0.11	0.0	0.04	0.08					
Log(RBC ratio)	7.05	1.90	4.37	6.82	7.34	6.92	0.64	6.36	6.53	7.13	7.1	1.22	6.25	6.81	7.47					
Leverage	0.6	0.25	0.45	0.62	0.76	0.78	0.23	0.7	0.88	0.94	0.52	0.21	0.41	0.57	0.68					
Cash-to-asset	0.02	0.06	0.0	0.01	0.03	0.02	0.05	0.0	0.0	0.01	0.03	0.06	0.0	0.01	0.04					
Bond portfolio rating	1.26	0.22	1.11	1.21	1.39	1.41	0.21	1.27	1.42	1.53	1.21	0.19	1.08	1.16	1.27					
VIF	0.14	0.38	0.02	0.04	0.1	0.05	0.07	0.01	0.02	0.06	0.18	0.44	0.03	0.05	0.12					
High VIF	0.07	0.12	0.01	0.03	0.07	0.03	0.05	0.0	0.01	0.04	0.08	0.13	0.02	0.04	0.08					
ExpW VIF	0.15	0.43	0.02	0.04	0.1	0.05	0.08	0.01	0.02	0.06	0.19	0.52	0.03	0.05	0.12					
FD VIF	0.21	0.55	0.03	0.06	0.15	0.07	0.1	0.01	0.02	0.08	0.26	0.64	0.04	0.08	0.18					
ExpW semi FD VIF	0.2	0.79	0.02	0.04	0.1	0.06	0.12	0.01	0.02	0.05	0.25	0.92	0.03	0.05	0.11					
Exposure to MF holdings	0.16	0.52	0.02	0.04	0.1	0.05	0.08	0.01	0.02	0.06	0.21	0.61	0.03	0.05	0.12					
Exposure to high impaired MF outflows	4.93	7.47	0.0	0.0	11.14	4.14	7.70	0.0	0.0	13.81	4.46	7.33	0.0	0.0	9.35					
Precrisis net buyers	0.12	0.27	0.0	0.0	0.0	0.16	0.29	0.0	0.0	0.23	0.11	0.26	0.0	0.0	0.0					
Crisis net buyers	0.24	0.41	0.0	0.0	0.0	0.32	0.47	0.0	0.0	1.0	0.21	0.41	0.0	0.0	0.0					
Precrisis net buyers	0.35	0.48	0.0	0.0	1.0	0.43	0.5	0.0	0.0	1.0	0.31	0.46	0.0	0.0	1.0					
Crisis net buyers	0.47	0.75	0.0	0.0	0.0	0.97	0.63	0.0	0.0	5.97	0.27	4.34	0.0	0.0	0.0					
Crisis net buyers	1.79	4.7	0.0	0.0	6.22	2.16	5.83	0.0	0.0	7.41	1.65	4.16	0.0	0.0	5.33					

This table reports summary statistics for the insurance companies and their aggregate trading in the *Precrisis* period (2/21/2020 to 3/6/2020) and the *Crisis* period (3/6/2020 to 3/19/2020). Our sample consists of 1,647 insurers, of which 465 are life insurers and 1,182 are P&C insurers, resulting in 3,294 insurer-period observations underlying this table. The column *All insurers* reports summary statistics for the full sample, while the *Life* and *Property & Casualty* columns present type-specific statistics. While insurers' aggregate trading variables are calculated using NAIC data, we obtain quarterly balance sheet data from S&P Global Market Intelligence, primarily for 2019Q4, unless otherwise noted as spanning the previous five years, for insurers' control variables. The insurers are compared across the following metrics: *Log(total assets)*, the logarithm of an insurer's total assets in dollars. *Asset growth*, the 5-year compound annual growth rate of total assets, expressed as a decimal fraction. *Log(RBC ratio)*, the logarithm of the Authorized Control Level (ACL) risk-based capital ratio expressed as a percentage. *Leverage*, defined as 1 minus the ratio of equity to total assets, expressed as a decimal fraction. *Cash-to-asset*, the ratio of an insurer's cash holdings to total assets, expressed as a decimal fraction. *Bond portfolio rating*, an insurer's weighted average NAIC bond portfolio rating (1=highest through 6=lowest). *Variation in insurance funding (VIF)* and *Variation in underwriting profitability (VUP)*, the standard deviation of an insurer's funding and underwriting profitability, respectively, over the past five years. *High VIF*, an indicator variable equal to 1 for insurers in the top quartile of the VIF distribution within each insurer type. *ExpW VIF*, the exponentially weighted standard deviation of an insurer's funding over the past 5 years. *FD VIF*, the standard deviation of an insurer's first-differenced funding over the past 5 years. *Semi VIF*, the semi-deviation of an insurer's funding over the past 5 years, separating only the negative funding fluctuations. *ExpW semi FD VIF*, the exponentially weighted semi-deviation of an insurer's first-differenced funding over the past 5 years. *Exposure to MF holdings*, the trade-weighted average exposure of an insurer to mutual fund holdings based on its bond trades where *MF holdings* is the natural logarithm of 1 plus the total par amount in dollars held by mutual funds in a given bond in a 2019Q4, calculated using eMax data. *Exposure to high impaired MF outflows*, the trade-weighted average exposure of an insurer to bonds with high impaired mutual fund outflows where *High impaired MF outflows* is an indicator variable equal to 1 if a bond's impaired mutual fund flow on a given day is negative and falls below the top quartile of negative flows (that is, the 80th percentile of outflows) and 0 otherwise. This variable is calculated using CRSP and Morningstar data. *Precrisis (crisis) net buyers*, the share of net buyers among insurers in the *precrisis (crisis)* period. *Precrisis (crisis) net buyers*, the log-modulus transformation of an insurer's aggregate net purchases in a given period, $\log(|\text{NetPurchases}_i|)$ where $i \in \{\text{Precrisis}, \text{Crisis}\}$.

Table 4
Variations in insurer funding and insurer trading during the COVID-19 crisis (additional robustness)

	(1)		(2)		(3)		(4)		(5)		(6)		(7)	
	Net buyer (diff. precrisis)	Net buyer (mutating bonds)	Net buyer (SE insurer group)	Net buyer (WLS)	Net buyer	Net buyer	Net buyer	Net buyer	Net buyer	Net buyer	Net buyer	Net buyer	Net buyer	Net buyer
VIF	-0.00078 (0.0179)	-0.0115 (0.0180)	-0.0106 (0.0228)	-0.0111 (0.0203)	-0.0115 (0.0231)	-0.0115 (0.0231)	-0.0115 (0.0231)	-0.0115 (0.0231)	-0.0115 (0.0231)	-0.0115 (0.0231)	-0.0115 (0.0231)	-0.0115 (0.0231)	-0.0115 (0.0231)	-0.0115 (0.0231)
Crisis + VIF	-0.00078** (0.0228)	-0.0115** (0.0203)	-0.0106** (0.0254)	-0.0111** (0.0232)	-0.0115** (0.0232)	-0.0115** (0.0232)	-0.0115** (0.0232)	-0.0115** (0.0232)	-0.0115** (0.0232)	-0.0115** (0.0232)	-0.0115** (0.0232)	-0.0115** (0.0232)	-0.0115** (0.0232)	-0.0115** (0.0232)
Insurer-period controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Net	3.284	3.284	3.284	3.284	3.284	3.284	3.284	3.284	3.284	3.284	3.284	3.284	3.284	3.284
Req	0.088	0.058	0.087	0.092	0.092	0.092	0.092	0.092	0.092	0.092	0.092	0.092	0.092	0.092

B. Exponentially weighted semi-deviation of first-differences in insurance funding

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Net buyer (diff. precrisis)	Net buyer (mutating bonds)	Net buyer (SE insurer group)	Net buyer (WLS)	Net buyer	Net buyer	Net buyer
ExpW semi FD VIF	0.00068 (0.0118)	-0.0122 (0.00928)	-0.0114 (0.0118)	-0.0111 (0.0109)	-0.0107 (0.0113)	0.00077 (0.0109)	0.0118 (0.0281)
Crisis + expW semi FD VIF	-0.0056** (0.0192)	-0.0423** (0.0146)	-0.0482** (0.0188)	-0.0478** (0.0166)	-0.0478** (0.0166)	-0.0482** (0.0166)	-0.0482** (0.0166)
Insurer-period controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Net	3.284	3.284	3.284	3.284	3.284	3.284	3.284
Req	0.088	0.058	0.087	0.092	0.092	0.092	0.092

The sample is constructed at the insurer-period level, aggregating an insurer's net purchases—defined as total buys minus total sells—in the *Precrisis* period (2/21/2020 to 3/6/2020) and the *Crisis* period (3/6/2020 to 3/19/2020). In columns 1–4, the dependent variable, *Net Buyer*, is an indicator variable equal to 1 if an insurer's aggregate net purchases in a given period are positive and 0 otherwise. Column 1 accounts for an alternative *Precrisis* period (1/2/2020 to 1/15/2020) that reflects stable market conditions and precedes the broader market reaction to the pandemic. Column 2 adjusts an insurer's net purchases for maturing bonds before computing *Net Buyer*. Column 3 uses standard errors clustered at the insurer group level. In column 5, the dependent variable, *Net Buyer*, is the log-modulus transformation of an insurer's aggregate net purchases in a given period, $\log(|\text{NetPurchases}_i|)$ where $i \in \{\text{Precrisis}, \text{Crisis}\}$. In columns 6 and 7, the dependent variables—*Net Buyer Over Assets* and *Net Buyer Over Bonds*, respectively—are defined as an insurer's aggregate net purchases in a given period, scaled by either total assets or total bond holdings as of 2019Q4 and given in percentages. These trading variables are calculated using NAIC and S&P Global Market Intelligence data. *Crisis* is an indicator variable that equals 1 on or after 3/6/2020. Other independent variables are drawn from quarterly balance sheet data from S&P Global Market Intelligence, primarily for 2019Q4, unless otherwise noted as spanning the past five years. Panel A focuses on *Variation in insurance funding (VIF)*, which is defined as the standard deviation of an insurer's funding over the past 5 years. Panel B focuses on *ExpW semi FD VIF*, which is defined as the exponentially weighted semi-deviation of an insurer's first-differenced funding over the past 5 years. *Insurer-period controls*, defined as the control variables interacted with a period indicator variable, are specified as follows but omitted from Table 4 (for brevity: *Log(total assets)* is the logarithm of an insurer's total assets. *Asset growth* is the 5-year compound annual growth rate of total assets. *Log(RBC ratio)* is the logarithm of the ACL risk-based capital ratio. *Leverage* is defined as 1 minus the ratio of equity to total assets. *Cash-to-asset* is the ratio of an insurer's cash holdings to total assets. *Bond portfolio rating* is an insurer's weighted average NAIC bond portfolio rating (1=highest through 6=lowest). *Insurer type fixed effects* differentiate life and P&C insurers. *Period fixed effects* differentiate the *precrisis* and *crisis* periods. Unless otherwise stated, standard errors are clustered at the insurer level and reported in parentheses. *p < .1. **p < .05. ***p < .01.

Read:

- Table 2 describes insurer characteristics before the COVID-19 crisis. Life insurers are larger and more leveraged, while P&C insurers are more numerous.
- The table also shows variation across insurers in funding stability, which is essential for the cross-sectional tests.

- Table 3 is the first main regression table. The crisis interaction with VIF is negative and significant: insurers with more stable funding were more likely to be net buyers during the crisis.
- Economically, a one-standard-deviation decrease in VIF increases the probability of being a crisis-period net buyer by about 3.1 percentage points.
- Table 4 shows robustness: the result survives alternative precrisis windows, maturing-bond adjustments, insurer-group clustering, size-weighting, alternative dependent variables, and alternative funding-stability measures.
- The volume result is also economically large: a one-standard-deviation decrease in funding variation raises crisis-period net purchase volume by about 16%–19%, depending on the specification.

Table 3
Variations in insurer funding and insurer trading during the COVID-19 crisis

	(1) Net buyer	(2) Net buyer	(3) Net buyer	(4) Net buyer	(5) Net buyer	(6) Net buyer	(7) Net buyer	(8) Net buyer
VIF	-0.087*** (0.0162)	-0.106 (0.0197)						
Crisis × VIF	-0.0642*** (0.0160)	-0.081*** (0.0219)						
VIUP			-0.0500 (0.0207)					
Crisis × VIUP			-0.172** (0.0746)					
High VIF				0.0317 (0.0246)				
Crisis × high VIF				-0.113*** (0.0227)				
ExpW VIF					-0.0108 (0.0152)			
Crisis × ExpW VIF					-0.064*** (0.0172)			
FD VIF						-0.00641 (0.0143)		
Crisis × FD VIF						-0.0549*** (0.0159)		
Semi VIF							-0.00514 (0.00913)	
Crisis × semi VIF							-0.0274** (0.0096)	
ExpW semi FD VIF								-0.0134 (0.0104)
Crisis × expW semi FD VIF								-0.0482*** (0.0209)**
logtotal assets	0.0520*** (0.00648)	0.0518*** (0.00647)	0.0534*** (0.00653)	0.0520*** (0.00648)	0.0520*** (0.00648)	0.0520*** (0.00648)	0.0520*** (0.00648)	0.0520*** (0.00648)

Continued

Table 3
(Continued)

	(1) Net buyer	(2) Net buyer	(3) Net buyer	(4) Net buyer	(5) Net buyer	(6) Net buyer	(7) Net buyer	(8) Net buyer
Crisis × logtotal assets								
Asset growth								
Crisis × asset growth								
log(RBC ratio)								
Crisis × log(RBC ratio)								
Leverage								
Crisis × leverage								
Cash-to-assets								
Crisis × cash-to-assets								
Bond portfolio rating								
Crisis × bond portfolio rating								
Insurer type-period fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Notes	3,204	3,204	3,204	3,204	3,204	3,204	3,204	3,204
Ref	0.056	0.087	0.088	0.087	0.087	0.087	0.086	0.087

The sample is constructed at the insurer-period level, aggregating an insurer's net purchases—defined as total buys minus total sells—in the Pre-crisis period (2/1/2020 to 3/31/2020) and the Crisis period (3/31/2020 to 3/31/2021). The dependent variable, *Net Buyer*, is an indicator variable equal to 1 if an insurer's aggregate net purchases in a given period are positive and 0 otherwise. These trading variables are calculated using NAIC data. Column 1 includes only insurer type-period fixed effects. Columns 2–8 include the full set of controls. *Crisis* is an indicator variable that equals 1 in or after 3/31/2020. Other independent variables are drawn from quarterly balance sheet data from S&P Global Market Intelligence, primarily for 2019Q4, unless otherwise noted as spanning the past 5 years. *Variation in insurance funding (VIF)* and *variation in underwriting profitability (VIUP)* are the standard deviations of an insurer's funding and underwriting profitability, respectively, over the past 5 years. *High VIF* is an indicator variable equal to 1 for insurers in the top tercile of the VIF distribution within each insurer type. *ExpW VIF* is the exponentially weighted standard deviation of an insurer's funding over the past 5 years. *FD VIF* is the standard deviation of an insurer's first-differenced funding over the past 5 years. *Semi VIF* is the semi-deviation of an insurer's funding over the past 5 years. *logtotal assets* is the logarithm of an insurer's total assets. *Asset growth* is the 5-year compound annual growth rate of total assets. *log(RBC ratio)* is the logarithm of the RBC risk-based capital ratio. *Leverage* is defined as 1 minus the ratio of equity to total assets. *Cash-to-assets* is the ratio of an insurer's cash holdings to total assets. *Bond portfolio rating* is an insurer's weighted average NAIC bond portfolio rating (1=highest through 4=lowest). *Insurer type-period fixed effects* differentiate life and P&C insurers. *Period fixed effects* differentiate the precrisis and crisis periods. Standard errors are clustered at the insurer level and reported in parentheses. Notes and Ref refer to the number of observations and R-squared, respectively. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.3 Table 5 and Table 6: relationship trading and dealer reciprocity

Read:

- Table 5 asks which dealers insurers supported. The answer is relationship dealers.
- In the crisis, insurers bought more from dealers with whom they had stronger prior relationships and sold less to those dealers.
- This buy-sell asymmetry is important because it shows targeted liquidity support, not just routine relationship trading.
- The result survives insurer-day and dealer-day fixed effects and tests that exclude underwriter-dealer relationships.
- Table 6 asks whether dealers later rewarded supporting insurers. They did: in the postcrisis period, supporting insurers bought at prices about 15 basis points lower and sold at prices about 28 basis points higher than other insurers, conditional on dealer, bond, day, and trade size.
- The relationship is therefore mutually beneficial: insurers help dealers during stress, and dealers provide better execution later.

5.4 Table 7 and Table 8: stable-funded insurer relationships and transaction costs

Read:

	(1) Share of insurer i's buys from dealer j	(2) Share of insurer i's sells to dealer j	(3) Share of insurer i's buys from dealer j	(4) Share of insurer i's sells to dealer j	(5) Share of insurer i's buys from dealer j	(6) Share of insurer i's sells to dealer j	(7) Share of insurer i's buys from dealer j	(8) Share of insurer i's sells to dealer j
Crisis	0.0115 (0.0230)	-0.0347* (0.0183)	-0.182*** (0.0251)	-0.0099*** (0.0176)				
Share of past trading	2.334*** (0.334)	2.637*** (0.300)	3.927*** (0.311)	2.279*** (0.219)	3.835*** (0.318)	2.314*** (0.264)	2.988*** (0.297)	1.839*** (0.233)
Crisis × share of past trading	2.055*** (0.437)	-1.176** (0.355)	1.222*** (0.254)	-0.662*** (0.194)	1.362*** (0.306)	-0.706** (0.301)	1.128** (0.483)	-0.482* (0.289)
log(total assets)	0.181*** (0.00812)	0.122*** (0.00635)	0.072*** (0.00337)	0.00777 (0.00826)				
Asset growth	0.311*** (0.0826)	0.0337 (0.0480)						
log (RBC ratio)	-0.0803*** (0.0110)	-0.0137* (0.00777)						
Leverage ratio	-0.649*** (0.0642)	-0.289*** (0.0461)						
Cash-to-assets	0.803*** (0.151)	0.271*** (0.0910)						
Avg. bond rating	0.643*** (0.0681)	0.559*** (0.0487)						
Insurer fixed effects	No	No	Yes	Yes	No	No	No	No
Dealer fixed effects	Yes	Yes	Yes	Yes	No	No	No	No
Day fixed effects	No	No	Yes	Yes	No	No	No	No
Insurer-day fixed effects	No	No	No	No	Yes	Yes	Yes	Yes
Dealer-day fixed effects	No	No	No	No	Yes	Yes	Yes	Yes
Nobs	706,540	706,540	887,304	887,304	878,218	878,218	867,192	867,192
Rsq	0.016	0.019	0.058	0.060	0.079	0.083	0.072	0.066

The sample is constructed at the insurer-dealer-day level, spanning the *Pre-crisis* period (2/21/2020 to 3/6/2020) and the *Crisis* period (3/6/2020 to 3/19/2020). All of the insurer-dealer pairs that traded at least once between 2017 and 2019 are considered potential pairings and are included in the sample. The dependent variables in columns 1 through 8 are defined as follows: *Share of insurer i's buys from (sells to) dealer j* refers to the ratio of the total net amount of bonds that insurer i bought from (sold to) dealer j over the total net amount traded by insurer i on a given day. These trading variables are calculated using NAIC data. Columns 7 and 8 exclude dealers with underwriter commitments. The independent variables are defined as follows: *Crisis* is an indicator variable that equals 1 on or after 3/6/2020. *Share of past trading* is the ratio of the average monthly trade volume between insurer i and dealer j to insurer i's total trading between 2017 and 2019, calculated using NAIC data. Other independent variables are drawn from quarterly balance sheet data for 2019Q4 from S&P Global Market Intelligence. *log(total assets)* is the logarithm of an insurer's total assets. *Asset growth* is the 1-year compound annual growth rate of total assets. *log(RBC ratio)* is the logarithm of the ACL risk-based capital ratio. *Leverage* is defined as 1 minus the ratio of equity to total assets. *Cash-to-assets* is the ratio of an insurer's cash holdings to total assets. *Bond portfolio rating* is an insurer's weighted average NAIC bond portfolio rating (1=highest through 6=lowest). Standard errors are clustered at the day and the insurer-dealer levels and reported in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$.

	(1) Price difference (insurer buys)	(2) Price difference (insurer sells)	(3) Price difference (insurer buys)	(4) Price difference (insurer sells)
<i>A. Pre-crisis period</i>				
<i>Zero cutoff</i>				
Crisis net buyer	-0.152*** (0.0271)	0.2942*** (0.1297)		
Nobs	3,213	1,515		
<i>Dealer median cutoff</i>				
Above-median crisis trading	-0.1457* (0.0535)	0.2757*** (0.1140)		
Nobs	3,034	1,494		
<i>B. Pre-COVID-19 period</i>				
<i>Zero cutoff</i>				
Crisis net buyer			-0.0133 (0.0173)	0.0346 (0.0173)
Nobs			9,805	4,674
<i>Dealer median cutoff</i>				
Above-median crisis trading			-0.0322 (0.0174)	-0.0376 (0.0174)
Nobs			9,073	4,323

The sample is constructed at the insurer-dealer-trade level for two separate periods: Panel A covers the *Pre-crisis* period (4/1/2020 to 12/31/2020), while panel B covers the *Pre-COVID-19* period (1/1/2017 to 12/31/2019). The table presents differences in the transaction prices of trades by insurance companies that were net buyers versus net sellers during the *Crisis* period (3/6/2020 to 3/19/2020) after controlling for trade direction, bond, day, trade size, and dealer. These transaction price differences are calculated using NAIC data. On a given trading day, we separate the trade in a given bond executed by a given dealer into insurers' buys and sells. Within each subsample, we classify each trade into size categories based on its dollar transaction value: *micro* (\$1 to \$100,000), *odd lot* (\$100,000–\$1,000,000), *round lot* (\$1,000,000–\$5,000,000), and *block* (above \$5,000,000). For each bond, trading day, size category, and dealer, we then calculate the average transaction price (as a percentage of par). Within each panel, we differentiate between the *Zero cutoff* and the *Dealer median cutoff*. Under the *Zero cutoff*, we compare average prices for insurers that were net buyers versus net sellers during the *Crisis* period. Under the *Dealer median cutoff*, we compare average transaction prices for insurers that were above versus below a given dealer's median net trading volume with insurance companies during the *Crisis* period. To be included in the calculation, a bond must be traded at least once by each group of traders with the same dealer in the same size category on the same day. Standard errors are clustered at the bond and the day levels and reported in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$.

- Table 7 is the main liquidity-provision result. Dealers whose relationship insurers had less stable funding charged higher transaction costs.
- During the crisis, a one-standard-deviation reduction in connected-insurer VIF lowers transaction costs by about 13 basis points in the baseline specification.
- With bond-day-trade-size-trade-direction fixed effects, the reduction is about 9 basis points. This is a demanding comparison because it holds constant the traded bond, date, trade direction, and trade size category.
- The effect is stronger for dealer buys than dealer sells, consistent with the mechanism that dealers need stable investors when they absorb customer sales and accumulate inventory.
- Table 8 shows that the transaction-cost effect is mainly present in investment-grade bonds. This makes sense because insurers' mandates and risk-based capital rules make them more natural buyers of investment-grade credit than high-yield bonds.
- The effect is present across maturity groups and is not confined to tiny trades; it is stronger in larger trades than in small trades.

5.5 Figure 2 and Table 9: aggregate yield impact

Read:

- Table 9 shows that insurers buy more from issuers they already hold. Familiarity matters, especially when information acquisition is costly during a crisis.
- The yield counterfactual uses this evidence to ask how yields would have moved without ownership by stable-funded insurers.
- Figure 2 shows that stable-funded insurer ownership had little yield effect before the crisis but a large effect once the crisis began.
- For the median investment-grade corporate bond, the yield would have been about 25 basis points higher absent stable-funded insurer ownership.
- The effect is heterogeneous and clearer for investment-grade than high-yield bonds, again consistent with insurers' natural investment habitat.

Table 7
Insurer funding stability and bond liquidity during the COVID-19 crisis

	(1) Cost (all trades)	(2) Cost (all trades)	(3) Cost (dealer buys)	(4) Cost (dealer sells)	(5) Cost (all trades)
Crisis	26.18*** (2.823)				
VIF	0.125*** (0.0158)	0.120*** (0.0153)	0.0693*** (0.0154)	0.172*** (0.0169)	0.0986*** (0.0243)
Crisis * VIF	0.238*** (0.0360)	0.152*** (0.0287)	0.206*** (0.0423)	0.0927*** (0.0290)	0.445*** (0.0753)
Time to maturity	20.95*** (1.266)				
Age	1.399 (0.904)				
Outstanding amount	-3.394*** (1.276)				
Insurer trading share					-11.48*** (2.907)
Crisis * insurer trading share					-40.12*** (9.321)
Dealer market share					-0.989*** (0.179)
Crisis * dealer market share					-1.104 (0.979)
Underwriter					-2.793*** (1.074)
Crisis * underwriter					-3.072* (1.679)
Dealer size					-0.445 (0.923)
Crisis * dealer size					-6.002** (2.565)
Dealer leverage					-65.33*** (7.559)
Crisis * dealer leverage					123.7*** (22.65)
Dealer ROA					44.81*** (6.550)
Crisis * dealer ROAs					15.08 (15.25)
Rating fixed effects	Yes	No	No	No	No
Trade size fixed effects	Yes	No	No	No	No
Trade direction fixed effects	Yes	No	No	No	No
Bond-day-trade size-trade e direction fixed effects	No	Yes	Yes	Yes	Yes
Nobs	191,694	102,063	57,100	44,963	66,268
Rsq	0.165	0.650	0.632	0.679	0.698

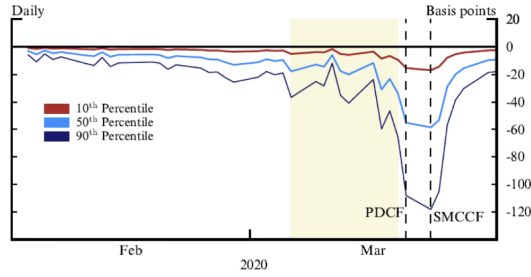
The sample is constructed at the trade level, spanning the *Pre-crisis* period (2/21/2020 to 3/6/2020) and the *Crisis* period (3/6/2020 to 3/19/2020), respectively. *Cost* refers to the transaction cost for a customer-dealer trade, estimated using Equation (5). While columns 1, 2, and 5 present estimates on *All trades*, columns 3 and 4 present estimates for *Dealer buys* and *Dealer sells*. These transaction cost variables are calculated using TRACE data. The independent variables are defined as follows: *Crisis* is an indicator variable that equals 1 on or after 3/6/2020. Here, *Variation in insurance funding (VIF)* refers to the average variation in insurance funding across a dealer's connected insurers, weighted by each insurer's total trading volume with the dealer from 2017 to 2019, calculated using NAIC and S&P Global Market Intelligence data. *Age* and *Time to maturity* refer to the logarithm of 1 plus the number of years since issuance and the number of years to maturity, respectively. *Outstanding amount* refers to the logarithm of 1 plus a bond's total par amount outstanding in thousands of dollars. *Insurer trade share* refers to the share of a dealer's trading volume with insurance companies relative to its total trading volume in 2019 in decimal fraction, calculated using NAIC and TRACE data. *Dealer market share* is the dealer's trading volume as a share of total market trading volume from 2017 and 2019, expressed in percentages, and it is calculated using the TRACE data. *Underwriter* is an indicator variable that equals 1 if a dealer is also a lead underwriter for the traded bond. *Dealer size* is the logarithm of a dealer's total assets. *Dealer leverage* is 1 minus the ratio of equity to total assets. *Dealer ROA* refers to a dealer's net income divided by total assets. Dealer balance sheet data are obtained from FOCUS reports. *Rating fixed effects* are based on each bond's composite rating. *Trade size fixed effects* are based on four size categories formed on the par amount traded: micro (\$1 to \$100,000), odd-lot (\$100,000 to \$1,000,000), round-lot (\$1,000,000 to \$5,000,000), and block (above \$5,000,000). *Trade direction fixed effects* differentiate dealer purchases and dealer sales. Standard errors are clustered at the day and the dealer-bond levels and reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 8
Insurer funding stability and bond liquidity during the COVID-19 crisis (by bond and trade characteristics)

	(1) Cost (IG bonds)	(2) Cost (HY bonds)	(3) Cost (≤ 7 years)	(4) Cost (> 7 years)	(5) Cost ($\leq \$100,000$)	(6) Cost ($> \$100,000$)
VIF	0.0875*** (0.0171)	0.213*** (0.0171)	0.0800*** (0.0118)	0.167*** (0.0223)	0.120*** (0.0134)	0.118*** (0.0220)
Crisis * VIF	0.189*** (0.0308)	0.189*** (0.0274)	0.173*** (0.0283)	0.131*** (0.0487)	0.129*** (0.0328)	0.203*** (0.0564)
Bond-day-trade size trade direction fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Nobs	79,285	22,778	66,512	35,451	67,538	34,545
Rsq	0.649	0.650	0.634	0.658	0.612	0.709

The sample is constructed at the trade level, spanning the *Pre-crisis* period (2/21/2020 to 3/6/2020) and the *Crisis* period (3/6/2020 to 3/19/2020). *Cost* refers to the transaction cost for a customer-dealer trade, estimated using Equation (5). The table presents estimates for all trades, without differentiating by trade direction. In columns 1-6, the table separates estimates for trades in investment-grade (IG) bonds, high-yield (HY) bonds, short & medium maturity (≤ 7 years) bonds, long maturity (> 7 years) bonds, small ($\leq \$100,000$) transaction size, and Large ($> \$100,000$) transaction size. These transaction cost variables are calculated using TRACE data. The independent variables are defined as follows: *Crisis* is an indicator variable that equals 1 on or after 3/6/2020. *Variation in insurance funding (VIF)* refers to the average variation in insurance funding across a dealer's connected insurers, weighted by each insurer's total trading volume with the dealer from 2017 to 2019, calculated using NAIC and S&P Global Market Intelligence data. Trade size categories in the *Bond-day-trade size fixed effects* are based on size categories formed on the par amount traded: micro (\$1 to \$100,000), odd-lot (\$100,000 to \$1,000,000), round-lot (\$1,000,000 to \$5,000,000), and block (above \$5,000,000). Standard errors are clustered at the day and the dealer-bond levels and reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

A Investment-grade corporate bond yields



B High-yield corporate bond yields

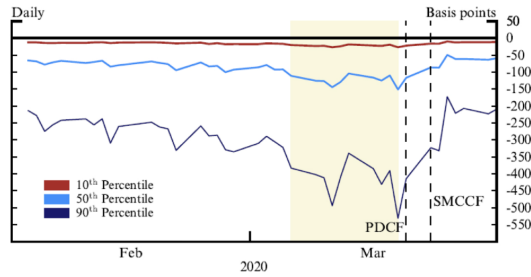


Figure 2
Effects of stable-funded insurers' bond issuer holdings on corporate bond yields during the COVID-19 crisis
Panel A shows the effect of stable-funded insurers' bond issuer holdings on investment-grade corporate bond yields from 2/1/2020 to 3/31/2020. Panel B presents the corresponding effect on high-yield corporate bond yields over the same period. The lines plot the differences between actual and counterfactual yields in the absence of stable-funded insurers' bond issuer holdings at the 10th, 50th, and 90th percentiles of holdings; see Equation (8). Bond yields are calculated using TRACE data, and insurers' bond issuer holdings are obtained from NAIC data. The shaded areas mark the crisis period from 3/6/2020 to 3/19/2020. Source: Authors' creation, using data from TRACE and NAIC.

Table 9
Insurers' net purchases and bond characteristics during the COVID-19 crisis

	(1) Net buys	(2) Net buys	(3) Net buys
Issuer holdings	0.294*** (0.0391)	0.362*** (0.0345)	0.392*** (0.0385)
Crisis * issuer holdings	0.0291 (0.0461)	-0.0581 (0.0430)	-0.0674 (0.0449)
Time to maturity	-0.0227 (0.252)		
Crisis * time to maturity	-0.309 (0.276)		
Age	-0.331*** (0.121)		
Crisis * age	0.248* (0.144)		
Outstanding amount	0.0371 (0.0932)		
Crisis * outstanding amount	-0.262*** (0.0953)		
Coupon	0.00109 (0.0380)		
Crisis * coupon	-0.0175 (0.0639)		
Rating-period fixed effects	Yes	No	No
Bond-period fixed effects	No	Yes	Yes
Insurer type-period fixed effects	No	No	Yes
Nobs	4,776	2,913	2,913
Rsq	0.227	0.574	0.577

The sample is constructed at the insurer-bond-period level, aggregating an insurer's net purchases in a given bond—defined as buys minus sells—in the *Pre-crisis* period (2/21/2020 to 3/6/2020) and the *Crisis* period (3/6/2020 to 3/19/2020), focusing only on those bonds an insurer bought on net. The dependent variable, *Net Buys*, is the logarithm of an insurer *i*'s aggregate net purchases, $\log(\text{Net purchases}_{i,b,t})$ with $\text{Net purchases}_{i,b,t} > 0$, in bond *b* by issuer *i* at period *t* (Pre-crisis, Crisis). This trading variable is calculated using NAIC data. The independent variables, based on NAIC and Mergent FISD data, are defined as follows: *Crisis* is an indicator variable that equals 1 on or after 3/6/2020. *Issuer holdings* refers to the logarithm of 1 plus an insurer's aggregate holdings in thousands of dollars in the issuer of the given bond in 2019:Q4. *Time to maturity* and *Age* refer to the logarithm of 1 plus the number of years to maturity and the number of years since issuance, respectively. *Outstanding amount* refers to the logarithm of 1 plus a bond's total par amount outstanding in thousands of dollars. *Coupon* refers to a bond's coupon rate. *Rating fixed effects* are based on each bond's composite rating. *Insurer type fixed effects* differentiate life and P&C insurers. *Period fixed effects* differentiate the pre-crisis and crisis periods. Standard errors are clustered at the insurer and the bond levels and reported in parentheses. **p* < .1; ***p* < .05; ****p* < .01.

5.6 Figure 3 and Table 10: insurer buying and mutual-fund exposure

Read:

- Figure 3 shows that insurer net purchases during the crisis are concentrated in bonds with larger mutual fund holdings.
- Table 10 formalizes this point. During the crisis, insurer buys increase in mutual fund holdings, while insurer sells do not.
- The result survives issuer-day fixed effects, meaning the paper compares bonds issued by the same firm and traded on the same day.
- Replacing mutual fund holdings with high imputed mutual fund outflows produces the same conclusion: insurers buy more of the bonds likely facing flow-induced fire sales.
- This evidence links insurer resiliency directly to mutual-fund fragility.

5.7 Table 11 and Table 12: funding stability meets mutual-fund fire sales

Read:

- Table 11 asks whether the funding-stability effect is stronger when insurers trade bonds exposed to mutual-fund selling pressure.
- The crisis interaction among VIF and mutual-fund exposure is negative for net-buyer outcomes: stable-funded insurers are especially likely to buy in bonds exposed to mutual-fund stress.

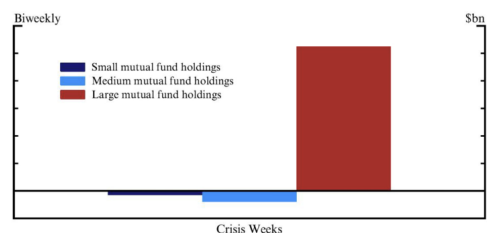


Figure 3
Insurer's net purchases and mutual fund bond holdings during the COVID-19 crisis
This figure presents insurers' net purchases in the secondary corporate bond market during the two-week crisis period, from 3/6/2020 to 3/19/2020. Net purchases are grouped into three categories based on mutual fund bond holdings as of 2019:Q4. Insurer net purchases are calculated using NAIC data, and mutual fund holdings are based on eMaxx data. *Source: Authors' creation, using data from NAIC and eMaxx.*

- Table 12 asks whether dealer transaction costs reflect the same mechanism.
- The triple interaction is positive: during the crisis, transaction costs are higher when dealer connected-insurer VIF is higher and the bond has greater mutual-fund exposure.
- Since lower VIF means more stable funding, this means stable-funded insurer relationships reduce transaction costs most in bonds facing the greatest mutual-fund selling pressure.
- Together, Tables 11 and 12 complete the paper's central mechanism: mutual funds create fire-sale pressure, stable-funded insurers absorb it, and dealers with access to those insurers provide better liquidity.

Table 11
Variations in insurer funding, insurer trading, and mutual fund holdings and flows during the COVID-19 crisis

	(1) Net buyer	(2) Net buyer	(3) Net buys	(4) Net buys
Exposure to MF holdings	0.0291 (0.0248)		0.0000522 (0.329)	
Crisis * exposure to MF holdings	0.0543* (0.0325)		0.544 (0.411)	
VIF	-0.0211 (0.0188)	-0.00843 (0.0175)	-0.166 (0.132)	0.0629 (0.171)
Crisis * VIF	-0.00398 (0.0208)	-0.0365* (0.0205)	0.100 (0.163)	-0.391* (0.231)
Exposure to MF holdings * VIF	0.00139 (0.0042)		0.0335 (0.0535)	
Crisis * exposure to MF holdings * VIF	-0.0183*** (0.0049)		-0.286*** (0.0689)	
Exposure to high imputed MF outflows		-0.506 (1.303)		-15.51 (16.82)
Crisis * exposure to high imputed MF outflows		2.610* (1.368)		33.21* (17.31)
Exposure to high imputed MF outflows * VIF		0.382 (0.590)		-5.941 (7.867)
Crisis * exposure to high imputed MF outflows * VIF		-0.864** (0.386)		-1.974 (5.831)
Insurer-period-exposure controls	Yes	Yes	Yes	Yes
Observations	3,294	3,294	3,294	3,294
R-squared	0.255	0.215	0.091	0.073

The sample is constructed at the insurer-period level, aggregating an insurer's net purchases—defined as total buys minus total sells—in the *Precrisis* period (2/21/2020 to 3/6/2020) and the *Crisis* period (3/6/2020 to 3/19/2020). In columns 1 and 2, the dependent variable, *Net Buyer*, is an indicator variable equal to 1 if an insurer's aggregate net purchases in a given period are positive and 0 otherwise. In columns 3 and 4, the dependent variable, *Net Buys*, is the log-modulus transformation of an insurer's aggregate net purchases in the period, $\text{sign}(\text{Net purchases}_i) \times \log(|\text{Net purchases}_i|)$ where $i \in (\text{Precrisis}, \text{Crisis})$. These trading variables are calculated using NAIC data. *Exposure to MF holdings* is the trade-weighted average exposure of an insurer to mutual fund holdings based on its bond trades in a given period. *MF holdings* is the natural logarithm of 1 plus the total par amount in dollars held by mutual funds in a given bond as of 2019:Q4, calculated using eMaxx data. *Exposure to high imputed MF outflows* represents the trade-weighted average exposure of an insurer to bonds with high imputed mutual fund outflows. *High imputed MF outflows* is an indicator variable equal to 1 if a bond's imputed mutual fund flow on a given day is negative and falls below the top quintile of negative flows (that is, the 80th percentile of outflows) and 0 otherwise. This variable is calculated using CRSP and Morningstar data. The other independent variables are drawn from quarterly balance sheet data from S&P Global Market Intelligence, primarily for 2019:Q4, unless otherwise noted spanning the past 5 years. *Variation in insurance funding (VIF)* is the standard deviations of an insurer's funding and underwriting profitability over the past 5 years. *Insurer-period-exposure controls*, defined as control variables interacted with a period indicator variable and the respective exposure measures, are specified as follows but omitted from Table 11 for brevity: *log(total assets)* is the logarithm of an insurer's total assets. *Asset growth* is the 5-year compound annual growth rate of total assets. *log(RBC ratio)* is the logarithm of the ACL risk-based capital ratio. *Leverage* is defined as one minus the ratio of equity to total assets. *Cash-to-assets* is the ratio of an insurer's cash holdings to total assets. *Bond portfolio rating* is an insurer's weighted average NAIC bond portfolio rating (1=highest through 6-lowest). *Insurer type fixed effects* differentiate life and P&C insurers. *Period fixed effects* differentiate the precrisis and crisis periods. Standard errors are clustered at the insurer level and reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 10
Insurer trading and mutual fund holdings and flows during the COVID-19 crisis

	(1) Insurer buys	(2) Insurer sells	(3) Insurer buys	(4) Insurer sells	(5) Insurer buys	(6) Insurer sells
MF holdings	-0.00513* (0.00274)	0.00368 (0.00427)	-0.00660 (0.00402)	0.00362 (0.00381)		
Crisis * MF holdings	0.0109** (0.00400)	0.00136 (0.00451)	0.0142** (0.00507)	-0.00413 (0.00486)		
Time to maturity	0.0023*** (0.0119)	-0.0104 (0.023)	0.0097*** (0.0152)	-0.0205 (0.028)	0.0097*** (0.0132)	-0.0193 (0.0196)
Age	-0.139* (0.0143)	-0.097** (0.023)	-0.146* (0.0166)	-0.0545* (0.0286)	-0.145** (0.0166)	-0.0559* (0.0281)
Outstanding amount	0.246*** (0.0185)	0.155*** (0.0200)	0.224*** (0.0234)	0.161*** (0.0177)	0.221*** (0.0223)	0.167*** (0.0156)
High imputed MF outflows					-0.0890** (0.0310)	0.0566 (0.0486)
Crisis * high imputed MF outflows					0.210*** (0.0581)	-0.0263 (0.0637)
Rating fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Day fixed effects	Yes	No	No	No	No	No
Insurer-day fixed effects	No	No	Yes	No	Yes	No
Nobs	90,850	90,850	84,031	84,031	84,031	84,031
Rsq	0.016	0.010	0.173	0.166	0.173	0.166

The sample is constructed at the bond-day level, spanning the *Precrisis* period (2/21/2020 to 3/6/2020) and the *Crisis* period (3/6/2020 to 3/19/2020). The dependent variables, *Insurer buys* and *Insurer sells*, refer to the logarithm of 1 plus the par amount of bonds bought and sold, respectively, by insurers for a given bond on a given day. These trading variables are calculated using NAIC data. The independent variables are defined as follows: *Crisis* is an indicator variable that equals 1 on or after 3/6/2020. *MF holdings* is the natural logarithm of 1 plus the total par amount in dollars held by mutual funds in a given bond as of 2019:Q4, calculated using eMaxx data. *High imputed MF outflows* is an indicator variable equal to 1 if a bond's imputed mutual fund flow on a given day is negative and falls below the top quintile of negative flows (that is, the 80th percentile of outflows) and 0 otherwise. This variable is calculated using CRSP and Morningstar data. *Time to maturity* and *Age* refer to the logarithm of 1 plus the number of years to maturity and the number of years since issuance, respectively. *Outstanding amount* refers to the logarithm of a bond's total par amount outstanding in thousands of dollars. *Rating fixed effects* are based on each bond's composite rating. Standard errors are clustered at the bond and the day levels and reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 12
Insurer funding stability, mutual fund ownership and flows, and bond liquidity during the COVID-19 crisis

	(1) Cost (all trades)	(2) Cost (all trades)	(3) Cost (all trades)	(4) Cost (all trades)
VIF	0.150*** (0.0287)		0.154*** (0.0300)	
Crisis * VIF	0.115*** (0.0354)		0.0920*** (0.0295)	
VIF * share by MF	0.0688** (0.0280)	0.0584*** (0.0196)		
Crisis * VIF * share by MF	0.153** (0.0601)	0.131** (0.0621)		
VIF * high imputed MF outflows			0.0354*** (0.0099)	0.0243*** (0.0081)
Crisis * VIF * high imputed MF outflows			0.0629** (0.0262)	0.0451* (0.0242)
Dealer market share	-0.800 (0.614)		-0.748 (0.615)	
Underwriter	-5.517*** (1.206)	-1.349* (0.732)	-5.609*** (1.200)	-1.441* (0.735)
Bond-day-trade size-trade direction fixed effects	Yes	Yes	Yes	Yes
Dealer-day fixed effects	No	Yes	No	Yes
Nobs	102,063	101,889	102,063	101,889
Rsq	0.65	0.71	0.65	0.71

The sample is constructed at the trade level, spanning the *Precrisis* period (2/21/2020 to 3/6/2020) and the *Crisis* period (3/6/2020 to 3/19/2020). *Cost* refers to the transaction cost for a customer-dealer trade, estimated using Equation (5). The table presents estimates for *All trades*, without differentiating by trade direction. These transaction cost variables are calculated using TRACE data. The independent variables are defined as follows: *Crisis* is an indicator variable that equals 1 on or after 3/6/2020. Here, *Variation in insurance funding (VIF)* refers to the average variation in insurance funding across a dealer's connected insurers, weighted by each insurer's total trading volume with the dealer from 2017 to 2019, calculated using NAIC and S&P Global Market Intelligence data. *Share by MF* is the percentage of a bond's amount outstanding that mutual funds hold as of 2019:Q4, which is calculated using eMaxx and Mergent FISD data. *High imputed MF outflows* is an indicator variable equal to 1 if a bond's imputed mutual fund flow on a given day is negative and falls below the top quintile of negative flows (that is, the 80th percentile of outflows) and 0 otherwise. This variable is calculated using CRSP and Morningstar data. *Dealer market share* is the dealer's trading volume as a share of total market trading volume from 2017 and 2019, and it is calculated using TRACE data. *Underwriter* is an indicator variable that equals 1 if a dealer is also a lead underwriter for the traded bond. Trade size categories in the *Bond-day-trade size fixed effects* are based on four size categories formed on the par amount traded: micro (\$1 to \$100,000), odd-lot (\$100,000 to \$1,000,000), round-lot (\$1,000,000 to \$5,000,000), and block (above \$5,000,000). The trade direction in the *Bond-day-trade size-trade direction fixed effects* differentiates dealer purchases and dealer sales. Standard errors are clustered at the day and the dealer-bond levels and reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

6 Short summary

- **Method.** The paper combines NAIC insurer bond transactions, regulatory TRACE dealer data, insurer statutory financial data, mutual fund holdings/flows, and dealer FOCUS reports to study liquidity provision during the COVID-19 corporate bond crisis.
- **Identification.** The key comparisons are across insurers with different funding stability, across insurer-dealer pairs with different prior relationships, across dealers connected to more or less stable-funded insurers, and across bonds with different mutual-fund fire-sale exposure.
- **Main result.** Insurers were important net buyers when dealers and mutual funds were under stress. More stable-funded insurers bought more, especially in bonds facing mutual-fund pressure.
- **Market mechanism.** Insurers used relationship dealers to provide liquidity, and dealers with stable-funded insurer networks charged lower transaction costs.
- **Takeaway.** Bond market resiliency depends on the liability structure of final investors, not only on dealer balance sheets or central-bank backstops.

7 Conclusion

7.1 Core conclusion

- The paper shows that insurance companies can act as stabilizing liquidity providers in corporate bond markets during periods of severe stress.
- Their stabilizing role is tied to funding stability: insurers with lower variation in insurance funding are more willing and able to buy when others sell.
- This insurer behavior improves dealer liquidity provision and reduces transaction costs, especially in bonds exposed to mutual-fund fire-sale pressure.

7.2 Economic intuition

- Dealers can intermediate ordinary imbalances, but they may not absorb a market-wide sell-off.
- Open-end mutual funds can amplify stress because redemption pressure forces sales of illiquid assets.
- Stable-funded insurers are different: they do not face the same immediate redemption pressure and can exploit temporary price dislocations.
- The asset-liability match matters. When the crisis does not directly damage insurers' funding or regulatory constraints, their long-term liabilities allow them to buy liquidity at a discount.

7.3 Takeaway

This paper shifts the discussion of bond market resiliency away from dealers alone. Dealer balance sheets matter, but the market also needs stable end investors who can absorb bonds during stress. O'Hara, Rapp, and Zhou show that insurers played this role in March 2020: they bought when

dealers and mutual funds were under pressure, supported relationship dealers, reduced transaction costs, and helped stabilize investment-grade corporate bond prices.